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Inventor(s)	Asakura; Hirotaka

Communication device, non-transitory computer-readable recording medium storing computer-readable instructions for communication device, and non-transitory computer-readable recording medium storing application program including computer-readable instructions for terminal device

Abstract

A communication device may: execute a Device Provisioning Protocol (DPP) process according to the DPP of a Wi-Fi standard, wherein the DPP process includes: receiving an authentication request in which a bootstrapping key of the communication device is used from a terminal device; in a case where the authentication request is received from the terminal device, sending an authentication response to the terminal device; and after the authentication response has been sent to the terminal device, receiving first connection information from the terminal device, and in a case where the first Wi-Fi connection between the communication device and the first access point is established, the communication device may send confirmation information to the terminal device, wherein the confirmation information is used by the terminal device for executing connectivity confirmation.

Inventors:	Asakura; Hirotaka (Nagoya, JP)
Applicant:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
Family ID:	1000008822163
Assignee:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11546755	12/2022	Jiang	N/A	H04L 9/0822
12041049	12/2023	Stevens	N/A	H04L 63/0876
2017/0257819	12/2016	McCann	N/A	H04L 67/56
2019/0303071	12/2018	Tsuji	N/A	G06F 3/1222
2019/0306707	12/2018	Shibata	N/A	H04W 12/0431
2019/0306710	12/2018	Cammarota	N/A	H04W 12/06
2020/0104081	12/2019	Miyake	N/A	G06K 7/10722
2020/0214059	12/2019	Suzuki	N/A	H04W 76/14
2020/0226258	12/2019	Nix	N/A	H04W 4/60
2021/0243825	12/2020	Takada	N/A	H04W 76/32
2021/0385778	12/2020	Ahn	N/A	H04W 60/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020-108118	12/2019	JP	N/A
7279367	12/2022	JP	H04L 9/0841

WO-2018203512	12/2017	WO	B62B 5/0096
WO-2019245190	12/2018	WO	N/A
WO-2020043809	12/2019	WO	H04L 63/0823

OTHER PUBLICATIONS

Wi-Fi Easy Connect Specification Version 2.0, Wi-Fi Alliance, 2020, Cited in the present application. cited by applicant

Wi-Fi Direct Specification Version 1.9, Wi-Fi Alliance, 2021, Cited in the present application. cited by applicant

Primary Examiner: Patel; Mahendra R

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Japanese Patent Application No. 2022-044588 filed on Mar. 18, 2022. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

(2) A technique to establish a Wi-Fi connection between a device and an access point by using a terminal device according to the Device Provisioning Protocol (DPP) of the Wi-Fi standard is known. In particular, this technique describes a technique to notify the terminal device of a result of execution of the DPP.

DESCRIPTION

(3) The present disclosure provides a technique to enhance user convenience when a Wi-Fi connection is to be established according to the DPP.

(4) A communication device is disclosed herein. The communication device may comprise: a Wi-Fi interface configured to execute Wi-Fi communication according to a Wi-Fi standard; and a controller. The controller may be configured to: execute a Device Provisioning Protocol (DPP) process according to the DPP of the Wi-Fi standard, wherein the DPP process comprises: receiving an authentication request in which a bootstrapping key of the communication device is used from a terminal device via the Wi-Fi interface; in a case where the authentication request is received from the terminal device, sending an authentication response to the terminal device via the Wi-Fi interface; and after the authentication response has been sent to the terminal device, receiving first connection information from the terminal device via the Wi-Fi interface, wherein the first connection information is used by the communication device for establishing a first Wi-Fi connection between the communication device and a first access point via the Wi-Fi interface, and in a case where the first Wi-Fi connection between the communication device and the first access point is established, send confirmation information to the terminal device via the Wi-Fi interface, wherein the confirmation information is used by the terminal device for executing connectivity confirmation to confirm whether the Wi-Fi communication between the communication device and the terminal device via the first access point is executable.

(5) According to the above configuration, the communication device sends the confirmation information to the terminal device in the case where the first Wi-Fi connection between the communication device and the first access point is established. Consequently, the terminal device

can execute connectivity confirmation by using the confirmation information. Thus, user convenience is enhanced.

(6) A non-transitory computer-readable recording medium storing an application program including computer-readable instructions for a terminal device is also disclosed herein. The terminal device may comprise: a Wi-Fi interface configured to execute Wi-Fi communication according to a Wi-Fi standard; and a processor. The computer-readable instructions, when executed by the processor, may cause the terminal device to: execute a Device Provisioning Protocol (DPP) process according to the DPP of the Wi-Fi standard, wherein the DPP process comprises: sending an authentication request in which a bootstrapping key of a communication device is used to the communication device via the Wi-Fi interface; in a case where the authentication request is sent to the communication device, receiving an authentication response from the communication device via the Wi-Fi interface; and after the authentication response has been received from the communication device, sending connection information to the communication device via the Wi-Fi interface, wherein the connection information is used by the communication device for establishing a first Wi-Fi connection between the communication device and an access point via the Wi-Fi interface, in a case where the first Wi-Fi connection between the communication device and the access point is established, receive confirmation information from the communication device via the Wi-Fi interface, wherein the confirmation information is used by the terminal device for executing connectivity confirmation to confirm whether the Wi-Fi communication between the communication device and the terminal device via the access point is executable; and in a case where the confirmation information is received from the communication device, execute the connectivity confirmation by using the confirmation information.

(7) According to the above configuration, the terminal device receives the confirmation information from the communication device in the case where the first Wi-Fi connection between the communication device and the access point is established. Consequently, the terminal device can execute the connectivity confirmation by using the confirmation information. Thus, the user convenience is enhanced.

Description

(1) A non-transitory computer-readable recording medium storing computer-readable instructions for the aforementioned communication device and a method implemented by the communication device are also novel and useful. Further, the terminal device implemented by this application program and a method executed by this terminal device are also novel and useful. Moreover, a communication system including the aforementioned communication device and the aforementioned terminal device is also novel and useful.

(2) FIG. 1 illustrates a configuration of a communication system.

(3) FIG. 2 is a sequence diagram of processes executed by devices.

(4) FIG. 3 is a sequence diagram of Case A.

(5) FIG. 4 is a sequence diagram continued from FIG. 3.

(6) FIG. 5 is a sequence diagram of Case B.

(7) FIG. 6 is a sequence diagram continued from FIG. 5.

(8) FIG. 7 is a sequence diagram of Case C.

(9) FIG. 8 is a sequence diagram of Case D.

(10) FIG. 9 is a sequence diagram of Case E.

(11) FIG. 10 is a sequence diagram of Case F.

(12) FIG. 11 is a sequence diagram of Case G.

FIRST EMBODIMENT

(13) (Configuration of Communication System 2; FIG. 1)

(14) As illustrated in FIG. 1, a communication system 2 includes two access points (AP) 6a, 6b, a printer 10 and a terminal 100. The present embodiment contemplates a situation in which a wireless connection according to the Wi-Fi standard (hereafter referred to as “Wi-Fi connection”) is to be established between the printer 10 and an AP (e.g., the AP 6a) by using the terminal 100.

(15) When a Wi-Fi connection between the terminal 100 and the AP 6a is established and a Wi-Fi connection between the printer 10 and the AP 6a is established, the terminal 100 and the printer 10 can communicate with each other via the AP 6a. However, there is a case where the terminal 100 and the printer 10 cannot communicate via the AP 6a even when the Wi-Fi connection between the devices 10, 100 and the AP 6a is established, due to, for example, weak radio environment around the AP 6a. To address this, in the present embodiment, connectivity confirmation to confirm whether the communication between the terminal 100 and the printer 10 via the AP 6a is executable is executed.

(16) (Configuration of Printer 10)

(17) The printer 10 is a peripheral device configured to execute a print function (e.g., a peripheral device of the terminal 100). In a modification, the printer 10 may be a multi-function device configured to execute a scan function, a facsimile function and the like as well as the print function. The printer 10 includes an operation unit 12, a display unit 14, a Wi-Fi interface 16, a print executing unit 18 and a controller 30. The units 12 to 30 are connected to bus lines (reference sign omitted). Hereafter, the interface will be referred to as “I/F”.

(18) The operation unit 12 includes a plurality of keys. The user can input various instructions to the printer 10 by operating the operation unit 12. The display unit 14 is a display to display various kinds of information. The display unit 14 is a touch screen, and also functions as the operation unit 12. The print executing unit 18 comprises a printing mechanism of an inkjet scheme, a laser scheme, or the like.

(19) The Wi-Fi I/F 16 is a wireless I/F to execute Wi-Fi communication according to the Wi-Fi standard. The Wi-Fi standard is a wireless communication standard to execute wireless communication according to, for example, 802.11 standard or its compliant standards (e.g., 802.11a, 11b, 11g, 11n, 11ac) of The Institute of Electrical and Electronics Engineers, Inc. (IEEE). The Wi-Fi I/F 16 in particular supports the Device Provisioning Protocol (DPP) created by the Wi-Fi alliance. Details of the DPP is described in the standard “Wi-Fi Easy Connect Specification Version 2.0” created by the Wi-Fi alliance.

(20) The Wi-Fi I/F 16 further supports Wi-Fi Direct (WFD) created by the Wi-Fi Alliance. Wi-Fi Direct is a registered trademark of Wi-Fi Alliance. Details of the WFD is described in the standard “Wi-Fi Direct Specification Version 1.9” created by the Wi-Fi alliance. In the WFD, a Group Owner state (hereafter referred to as “G/O state”) is defined. The G/O state is a state to operate as a parent station in a wireless connection established according to the WFD. Hereafter, the Wi-Fi connection established according to the WFD will be referred to as a “WFD connection”.

(21) The printer 10 can establish a Wi-Fi connection with the AP 6 according to an ordinary Wi-Fi scheme different from the DPP or the WFD (hereafter this may be referred to as “support the ordinary Wi-Fi scheme”). Specifically, in the ordinary Wi-Fi scheme, the printer 10 can use a Service Set Identifier (SSID) and a password of a wireless network formed by an AP (e.g., the AP 6a) to establish a Wi-Fi connection with this AP. Hereafter, the password will be referred to as “PW”.

(22) The Wi-Fi I/F 16 is physically one I/F (i.e., one IC chip). However, two MAC addresses MACpr1 and MACpr2 are assigned to the Wi-Fi I/F 16. The MAC address MACpr1 is a MAC address used for direct communication with a target device (i.e., communication not using another device). The MAC address MACpr2 is a MAC address used for communication with the target device via another device. The MAC address MACpr1 is used in for example the WFD. The MAC address MACpr2 is used in for example the DPP and the ordinary Wi-Fi scheme. Thus, the Wi-Fi I/F 16 may simultaneously execute both the WFD communication in which the MAC address

MACpr1 is used and the non-WFD communication in which the MAC address MACpr2 is used. In a modification, the MAC address MACpr1 may be used in the DPP as well as in the WFD. In another modification, the MAC address MACpr1 may be used in a SoftAP scheme instead of the WFD. The SoftAP scheme is a scheme for the printer **10** to operate as a parent station (i.e., SoftAP) of a Wi-Fi network to establish a Wi-Fi connection with another device (e.g., the terminal **100**).

(23) The controller **30** includes a CPU **32** and a memory **34**. The CPU **32** executes various processes according to a program **36** stored in the memory **34**. The memory **34** is configured of a volatile memory, a nonvolatile memory, and the like.

(24) The memory **34** further stores a public key PKpr, the above MAC addresses MACpr1, MACpr2 and a PIN code “1234”. Although not illustrated, the memory **34** stores a private key corresponding to the public key PKpr. The public key PKpr and the private key are used when authentication according to the DPP is executed. The PIN code “1234” is authentication information used when the WFD connection with the terminal **100** is established.

(25) A QR Code QR1 obtained by coding the public key PKpr, the MAC address MACpr2 and the PIN code “1234” is adhered to a housing **50** of the printer **10**. QR Code is registered trademark of DENSO WAVE INCORPORATED. Although the details will be described later, by the QR Code QR1 being captured, processes according to the DPP are executed between the printer **10** and the terminal **100**. In a modification, the QR Code QR1 may not be adhered to the housing **50** of the printer **10**. In this case, the printer **10** may be configured to display the QR Code QR1 on the display unit **14** when the printer **10** is powered on or when the printer **10** accepts a predetermined operation or a predetermined signal (e.g., an operation (or a signal) to cause the printer **10** to start a process to establish a Wi-Fi connection). As described above, the QR Code QR1 adhered to the housing **50** of the printer **10** (or the QR Code QR1 displayed on the display unit **14**) is obtained by coding the information including the PIN code “1234”. Thus, the user can establish the WFD connection between the printer **10** and the terminal **100** by capturing the QR Code QR1.

(26) (Configuration of Terminal **100**)

(27) The terminal **100** is a portable terminal device such as a cellphone (e.g., a smartphone), a PDA or a tablet PC. In a modification, the terminal **100** may for example be a desktop PC or a laptop PC. The terminal **100** includes an operation unit **112**, a display unit **114**, a Wi-Fi I/F **116**, a camera **118** and a controller **130**. The units **112** to **130** are connected to bus lines (reference sign omitted).

(28) The operation unit **112** is an I/F to input various instructions to the terminal **100**. The display unit **114** is a display to display various kinds of information. The display unit **114** is a so-called touch screen and also functions as the operation unit **112**. The camera **118** is a device to capture an object. In the present embodiment, the camera **118** is used to capture the QR Code of the AP6a and the QR Code of the printer **10**. The Wi-Fi I/F **116** is similar to the Wi-Fi I/F **16** of the printer **10**. In other words, the Wi-Fi I/F **116** supports the DPP, the WFD and the ordinary Wi-Fi scheme.

(29) The controller **130** includes a CPU **132** and a memory **134**. The CPU **132** executes various processes according to programs **136**, **138** stored in the memory **134**. The memory **134** is configured of a volatile memory, a nonvolatile memory, and the like.

(30) An Operating System (OS) program **136** is a program to implement basic operations of the terminal **100**. The application **138** is a program to establish a Wi-Fi connection between the terminal **100** and an AP (e.g., the AP **6a**) and establish a Wi-Fi connection between the printer **10** and the AP. Hereafter, the OS program and the application will be referred to as “OS” and “app”, respectively. The app **138** is installed in the terminal **100** from, for example, a server (not illustrated) on the Internet provided by a vendor of the printer **10** or from a server (not illustrated) on the Internet provided by a vendor of the OS **136**.

(31) The app **138** can cause the OS **136** to execute various processes and cause other app(s) to execute various processes via the OS **136** by supplying instructions to the OS **136**. Thus, the processes executed by the OS **136** or the other app(s) in response to the instructions from the app **138** can be said as processes implemented by the app **138** (i.e., the app **138** causes the CPU **132** to

function as an execution unit to execute these processes.)

(32) (Configurations of APs **6a**, **6b**)

(33) The AP **6a** is not a device in the G/O state of the WFD but is an ordinary AP called a wireless access point or a wireless LAN router. The AP **6a** supports the DPP scheme and the ordinary Wi-Fi scheme. The AP **6a** relays communication between a pair of devices which belong to a Wi-Fi network in which the AP **6a** operates as a parent station. The AP **6a** stores an SSID “S1” identifying this Wi-Fi network. The AP **6a** supports the DPP.

(34) Although not illustrated, the AP **6a** stores a public key PK_{apa} of the AP **6a**, a private key corresponding to the public key PK_{apa} and a MAC address of a Wi-Fi I/F of the AP **6a** (not illustrated). A QR Code obtained by coding information including the public key PK_{apa} and the MAC address of the AP **6a** is adhered to a housing of the AP **6a**.

(35) The AP **6b** has a similar configuration as the AP **6a** except that respective pieces of information (e.g., SSID, public key) are different.

(36) (Establishment of Wi-Fi Connection Between Terminal **100** and AP **6a** According to DPP Scheme; FIG. 2)

(37) Next, with reference to FIG. 2, a process to establish a Wi-Fi connection between the terminal **100** and the AP **6a** according to the DPP will be described. In the following descriptions in FIG. 2, for easier understanding, an action executed by CPUs of devices (e.g., the CPUs **32**, **132**) will be described with the devices (e.g., the printer **10**, the terminal **100**) as a subject of the action instead of describing the CPUs as the subject of the action. Further, all the communication executed by the printer **10** and the terminal **100** is executed via the Wi-Fi I/F **16** and the like. Accordingly, when communication is explained below, a phrase “via a Wi-Fi I/F” will be omitted. The same applies to explanations for FIG. 3 to FIG. 11.

(38) In T10, the terminal **100** accepts an operation to activate the app **138** from the user. Consequently, the app **138** is activated and the camera **118** is activated.

(39) In T12, the terminal **100** accepts an operation to capture the QR Code adhered to the housing of the AP **6a** by using the camera **118** from the user. In this case, the terminal **100** decodes the captured QR Code in T14, by which the terminal **100** obtains the public key PK_{apa} and the MAC address of the AP **6a**. The processes of T12 and T14 correspond to the Bootstrapping of the DPP.

(40) When the terminal **100** obtains the public key PK_{apa} and the MAC address in T14, the terminal **100** sends an Authentication Request in which the public key PK_{apa} is used to the AP **6a** in T20 with the MAC address as a recipient. Hereafter, Authentication will be referred to as “Auth” and Request will be referred to as “Req”.

(41) The Auth Req is a signal requesting for authentication of the terminal **100** which is a sender. Specifically, the terminal **100** firstly creates a shared key by using a private key of the terminal **100** (illustration omitted) and the public key PK_{apa} of the AP **6a** and creates encrypted data by encrypting a random value by using the shared key. The terminal **100** then sends the Auth Req including a public key of the terminal **100** (illustration omitted), the encrypted data and Capability of the terminal **100** to the AP **6a**. The Capability of the terminal **100** includes a value indicating that the terminal **100** can operate only as a Configurator of the DPP.

(42) When the AP **6a** receives the Auth Req from the terminal **100** in T20, the AP **6a** executes authentication of the encrypted data included in the Auth Req in T21. Specifically, the AP **6a** creates a shared key using the public key of the terminal **100** included in the Auth Req and a private key of the AP **6a** and decrypts the encrypted data by using the shared key. When decryption of the encrypted data succeeds, the AP **6a** determines that the authentication has succeeded and executes the processes from T22.

(43) In T22, the AP **6a** sends an Auth Response including a Capability of the AP **6a** to the terminal **100**. Hereafter, a Response will be referred to as “Res”. The Capability of the AP **6a** includes a value indicating that the AP **6a** can operate only as an Enrollee of the DPP.

(44) When the terminal **100** receives the Auth Res from the AP **6a** in T22, the terminal **100**

determines that the Capability of the AP **6a** included in the Auth Res (i.e., Enrollee) is not in conflict with the Capability of the terminal **100** itself (i.e., Configurator). The terminal **100** then sends an Auth Confirm in T23. The Auth Confirm includes information indicating that the terminal **100** is to operate as the Configurator and the AP **6a** is to operate as the Enrollee. Consequently, the terminal **100** determines to operate as the Configurator in T24. The Configurator is a device which plays a role to send a Configuration Object (hereafter referred to as “CO”) to be described to the Enrollee.

(45) In T26, the AP **6a** determines to operate as the Enrollee. The Enrollee is a device which plays a role to receive the CO from the Configurator. The processes of T20 to T26 correspond to the Auth of the DPP.

(46) In T30, the AP **6a** sends a Configuration Req to the terminal **100**. The Configuration will hereafter be referred to as “Config”. The Config Req is a signal requesting for sending of the CO.

(47) When the terminal **100** receives the Config Req from the AP **6a** in T30, the terminal **100** creates the CO for AP. Specifically, the terminal **100** firstly creates a Signed Connector (hereafter referred to as “SC”) for AP which is information to be used by the AP **6a** for establishing a Wi-Fi connection. The SC for AP includes for example a group ID which is an identifier identifying a wireless network. The terminal **100** then creates the CO for AP including the SC for AP and sends the Config Res including the CO for AP to the AP **6a** in T32.

(48) When the AP **6a** receives the Config Res from the terminal **100** in T32, the AP **6a** sends a Config Result to the terminal **100** in T34. The Config Result includes information indicating that the Config has succeeded. The processes of T30 to T34 correspond to the Config of the DPP. Hereafter, the Bootstrapping, the Auth and the Config of the DPP may collectively be referred to as “DPP process”.

(49) Next, the terminal **100** creates an SC for terminal which is information to be used by the terminal **100** for establishing a Wi-Fi connection. The SC for terminal includes a group ID which is the same as the group ID included in the SC for AP. Then, in T40, the terminal **100** sends a Discovery Req including the SC for terminal to the AP **6a**. The Discovery Req is a signal requesting for sending of a counterpart SC.

(50) When the AP **6a** receives the Discovery Req from the terminal **100** in T40, the AP **6a** executes authentication of the SC for terminal included in the Discovery Req by using the SC for AP. When the authentication of the SC for terminal succeeds, the AP **6a** creates a connection key. The AP **6a** then sends a Discovery Res including the SC for AP to the terminal **100** in T42.

(51) When the terminal **100** receives the Discovery Res from the AP **6a** in T42, the terminal **100** executes authentication of the SC for AP included in the Discovery Res by using the SC for terminal. When the authentication of the SC for AP succeeds, the terminal **100** creates a connection key. The connection key created at this point is the same as the connection key created by the AP **6a**. In other words, the connection key is shared by the terminal **100** and the AP **6a**. The processes of T40 and T42 correspond to the Network Access of the DPP.

(52) Next, the terminal **100** executes 4way-handshake communication with the AP **6a** by using the connection key. Consequently, in T50, a Wi-Fi connection between the terminal **100** and the AP **6a** is established. As a result of this, a wireless network in which the AP **6a** operates as a parent station is formed and the terminal **100** belongs to this wireless network as a child station.

(53) (Case A; FIG. 3 and FIG. 4)

(54) Next, with reference to FIG. 3 and FIG. 4, Case A in which a Wi-Fi connection between the printer **10** and the AP **6a** is established will be described. Case A is a case in which connectivity confirmation succeeds after the Wi-Fi connection between the printer **10** and the AP **6a** is established. In the initial state of FIG. 3, the processes according to the DPP have already been executed between the terminal **100** and the AP **6a** (see FIG. 2). As described above, the QR Code QR1 is adhered to the housing **50** of the printer **10**.

(55) When an operation to power on the printer **10** is executed by the user in T100, the printer **10** is

powered on in T102. In this case, the printer **10** causes the display unit **14** to display the QR Code QR1 in T104. This QR Code QR1 is the same as the QR Code QR1 adhered to the housing **50** of the printer **10**. In a modification, the process of T104 can be omitted.

(56) In T110, the terminal **100** accepts an operation to capture the QR Code QR1 (i.e., the QR Code adhered to the housing **50** of the printer **10** or the QR Code displayed in T104) by using the camera **118** from the user. In this case, the terminal **100** decodes the QR Code QR1 in T112, by which the terminal **100** obtains the public key PKpr, the MAC address MACpr2 and the PIN code “1234”.

(57) The processes of T120 to T134 are the same as the processes of T20 to T34 of FIG. 2 except that the communication is executed between the printer **10** and the terminal **100** and information used in the communication (e.g., the public key PKpr and the MAC address MACpr2) is different. In T132, a CO for printer is communicated instead of the CO for AP. The processes of T140 to T150 are the same as the processes of T40 to T50 of FIG. 2 except that communication is executed between the printer **10** and the AP **6a** and information used in the communication (e.g., an SC for printer) is different. Consequently, a Wi-Fi connection between the printer **10** and the AP **6a** is established.

(58) When the Wi-Fi connection between the printer **10** and the AP **6a** is established, an IP address IPpr is assigned from the AP **6a** to the printer **10**. The printer **10** receives the IP address IPpr from the AP **6a** in T152 and stores the received IP address IPpr in T154.

(59) In T156, the printer **10** sends a Status Query Result (hereafter referred to as “Result”) to the terminal **100**. The Result includes information indicating that the Wi-Fi connection between the printer **10** and the AP **6a** has been established.

(60) (Continuation of FIG. 3; FIG. 4)

(61) When the Wi-Fi connection is established in T150 of FIG. 3, the printer **10** shifts to the G/O state in T160 of FIG. 4. In the Wi-Fi network formed by the printer **10** which operates in the G/O state, the IP address IPpr assigned to the printer **10** (see T152, T154 of FIG. 3) is used as an SSID.

(62) When the terminal **100** receives the Result from the printer **10** in T156 of FIG. 3, the terminal **100** broadcasts a Probe Req in T162 of FIG. 4. The Probe Req is a signal to search for device(s) around the terminal **100**.

(63) When the printer **10** receives the Probe Req from the terminal **100** in T162, the printer **10** sends a Probe Res to the terminal **100** in T164. The Probe Res includes the SSID “IPpr”. As describes above, the printer **10** can send, to the terminal **100**, the IP address IPpr assigned from the AP **6a**.

(64) When the terminal **100** receives the Probe Res from the printer **10** in T164, the terminal **100** sends a connectivity confirmation signal which includes the IP address IPpr which matches the received SSID “IPpr” as a recipient address in T166. Consequently, the connectivity confirmation signal is sent from the terminal **100** to the printer **10** via the AP **6a**.

(65) The printer **10** receives the connectivity confirmation signal from the terminal **100** via the AP **6a** in T166. In this case, in T168, the printer **10** sends a connectivity OK signal indicating that the connectivity confirmation signal has been received to the terminal **100** via the AP **6a**.

(66) The terminal **100** receives the connectivity OK signal from the printer **10** via the AP **6a** in T168. In this case, the terminal **100** causes the display unit **114** to display a notification screen SC1 in T170. The notification screen SC1 includes a message indicating that communication with the printer **10** via the AP **6a** has succeeded. By viewing the notification screen SC1, the user can acknowledge that the terminal **100** and the printer **10** can communicate with each other via the AP **6a**.

(67) When a predetermined time elapses without the printer **10** receiving a WFD connection establishment request from the terminal **100** after the connectivity OK signal was sent to the terminal **100** in T168, the printer **10** stops the G/O state (i.e., the state in which the printer **10** operates as a parent station of the Wi-Fi network) in T172. Consequently, the printer **10** shifts to the state of not operating as a parent station of the Wi-Fi network and the Wi-Fi network formed by the

printer **10** disappears. Thus, processing load on the printer **10** can be reduced.

(68) (Case B; FIG. 5 and FIG. 6)

(69) Next, with reference to FIG. 5 and FIG. 6, Case B in which a Wi-Fi connection between the printer **10** and the AP **6a** is established will be described. Case B is a case in which the Wi-Fi connection between the printer **10** and the AP **6a** is established but the connectivity confirmation fails. FIG. 5 is continuation of FIG. 3. In other words, in the initial state of FIG. 5, the Wi-Fi connection between the printer **10** and the Ap **6a** is established.

(70) The processes of T260 to T264 are the same as the processes of T160 to T164 of FIG. 4. In T266, the terminal **100** sends the connectivity confirmation signal including the IP address IPpr as a recipient address to the printer **10**. In the present case, however, due to weak radio environment around the AP **6a**, the AP **6a** does not receive the connectivity confirmation signal. Consequently, the printer **10** does not receive the connectivity confirmation signal from the terminal **100**.

(71) Since the printer **10** does not receive the connectivity confirmation signal from the terminal **100**, the printer **10** does not send the connectivity OK signal. Consequently, the terminal **100** does not receive the connectivity OK signal from the printer **10**. When the terminal **100** does not receive the connectivity OK signal even after a predetermined time elapses since the connectivity confirmation signal was sent, the terminal **100** determines that the connectivity confirmation has failed and executes processes from T270. In a modification, a case in which the connectivity confirmation signal is suitably sent from the terminal **100** to the printer **10** but the connectivity OK signal is not sent from the printer **10** to the terminal **100** due to weak radio environment around the AP **6a** is also contemplated. In this case as well, the terminal **100** determines that the connectivity confirmation has failed and executes the processes from T270.

(72) In T270, the terminal **100** sends a WFD connection establishment request in which the obtained PIN code “1234” (see T112 of FIG. 3) is used to the printer **10**. Consequently, in T272, a WFD connection between the terminal **100** and the printer **10** is established. Consequently, the terminal **100** and the printer **10** can send and receive an inquiry and a response to be described later by using this WFD connection.

(73) In T274, the terminal **100** sends an inquiry to the printer **10** by using the established WFD connection. This inquiry is a signal requesting the printer **10** to send a response related to the connectivity confirmation.

(74) When the printer **10** receives the inquiry from the terminal **100** in T274, the printer **10** sends a response to this inquiry to the terminal **100** in T276 by using the WFD connection. In the present case, the printer **10** has not received the connectivity confirmation signal, thus this response includes information indicating that the connectivity confirmation signal has not been received. As described above, when the connectivity confirmation fails, the terminal **100** can send the inquiry and receive the response by using the WFD connection. In the aforementioned modification in which the AP **6a** does not receive the connectivity OK signal from the printer **10**, the response includes information indicating that the connectivity confirmation signal has already been received and the connectivity OK signal has already been sent. Although the details will be described later, when the Wi-Fi I/F **16** of the printer **10** malfunctions, the response further includes information indicating malfunction of the Wi-Fi I/F **16**.

(75) When the terminal **100** receives the response from the printer **10** in T276, the terminal **100** analyses the information in the response. In the present case, the response includes the information indicating that the connectivity confirmation signal has not been received and does not include the information indicating malfunction of the Wi-Fi I/F **16**. Thus, the terminal **100** determines that the Wi-Fi communication between the printer **10** and the terminal **100** via the AP **6a** cannot be executed because of the AP **6a**. In this case, the terminal **100** causes the display unit **114** to display a notification screen SC2 in T278. The notification screen SC2 includes a message indicating that communication with the printer **10** via the AP **6a** has failed and a message indicating a method to connect to another AP (e.g., the AP **6b**). By viewing the notification screen SC2, the user can

acknowledge what sort of approach should be taken to execute the Wi-Fi communication with the printer **10** (i.e., to connect to the other AP).

(76) (Continuation of FIG. 5; FIG. 6)

(77) In T280 of FIG. 6, the terminal **100** accepts input of an SSID “S2” and a PW “P2” of the AP **6b** from the user. In this case, the terminal **100** firstly disconnects the Wi-Fi connection with the AP **6a** in T282. Next, the terminal **100** sends a Wi-Fi connection establishment request in which the SSID “S2” and the PW “P2” are used to the AP **6b** in T284. Consequently, in T286, a Wi-Fi connection between the terminal **100** and the AP **6b** is established. Next, in T290, the terminal **100** sends the SSID “S2” and the PW “P2” to the printer **10** by using the WFD connection (see T272 of FIG. 5).

(78) When the printer **10** receives the SSID “S2” and the PW “P2” from the terminal **100** in T290, the printer **10** sends a Wi-Fi connection establishment request including the SSID “S2” and the PW “P2” to the AP **6b** in T292. Consequently, in T294, a Wi-Fi connection between the printer **10** and the AP **6b** is established. As a result of this, both the terminal **100** and the printer **10** belong to the Wi-Fi network formed by the AP **6b**. Thus, the terminal **100** and the printer **10** can communicate with each other via the AP **6b**.

(79) When the printer **10** establishes the Wi-Fi connection with the AP **6b** in T294, the printer **10** sends, to the terminal **100**, an establishment notification indicating that the Wi-Fi connection with the AP **6b** has been established by using the WFD connection in T296. Although the illustration is omitted, when the Wi-Fi connection between the printer **10** and the AP **6b** is established in T294, an IP address is assigned to the printer **10** from the AP **6b**. The establishment notification may include this IP address. Then, the terminal **100** may execute the connectivity confirmation (i.e., the same processes as T166 and T168 of FIG. 4) by using this IP address. When the terminal **100** receives the establishment notification from the printer **10** in T296, the terminal **100** sends a G/O stop instruction to the printer **10** by using the WFD connection in T298. Consequently, the printer **10** stops the G/O state in T299. Consequently, the Wi-Fi network formed by the printer **10** disappears. Thus, process load on the printer **10** can be reduced.

(80) (Case C; FIG. 7)

(81) Next, Case C will be described with reference to FIG. 7. In Case C, the cause of failure of the connectivity confirmation is different from Case B. Specifically, Case C is a case in which the connectivity confirmation fails due to malfunction of the Wi-Fi I/F **16** of the printer **10**. FIG. 7 is continuation of FIG. 3. In other words, in the initial state of FIG. 7, the Wi-Fi connection between the printer **10** and the AP **6a** is established. The processes of T360 to T374 are the same as the processes of T260 to T274 of FIG. 5.

(82) When the printer **10** receives the inquiry from the terminal **100** in T374, the printer **10** sends a response to this inquiry to the terminal **100** in T376 by using the WFD connection. In the present case, the printer **10** has not received the connectivity confirmation signal, thus the response includes information indicating that the connectivity confirmation signal has not been received yet. Moreover, since the Wi-Fi I/F **16** of the printer **10** malfunctions, this response further includes the information indicating malfunction of the Wi-Fi I/F **16**. In a modification, this response may include only the information indicating malfunction of the Wi-Fi I/F **16**. The processes of T380 and T382 are the same as the processes of T298 and T299 of FIG. 6.

(83) When the terminal **100** receives the response from the printer **10** in T376, the terminal **100** analyses the information in the response. In the present case, the response includes the information indicating malfunction of the Wi-fi I/F **16**. Thus, the terminal **100** determines that the Wi-Fi communication between the printer **10** and the terminal **100** via the AP **6a** cannot be executed because of the Wi-Fi I/F **16** of the printer **10**. In this case, in T384, the terminal **100** causes the display unit **114** to display a notification screen SC3. The notification screen SC3 includes a message indicating that communication with the printer **10** via the AP **6a** has failed and a message indicating that the user needs to contact a support center. The user can acknowledge what sort of

approach should be taken to execute Wi-Fi communication with the printer **10** (i.e., contact the support center) by viewing the notification screen SC3.

(84) (Case D; FIG. 8)

(85) Next, with reference to FIG. 8, Case D will be described. Case D is a case in which the DPP process between the terminal **100** and the printer **10** fails. The initial state of FIG. 8 is the same as the initial state of FIG. 3. The processes of T400 to T421 are the same as the processes of T100 to T121 of FIG. 3.

(86) In T420, the terminal **100** sends, to the printer **10**, the Auth Req including a value indicating that the terminal **100** can operate only as the Configurator of the DPP (see T20 of FIG. 2). In the present Case, also the printer **10** can operate only as the Configurator. Thus, the printer **10** determines that the roles of the printer **10** and the terminal **100** are in conflict, and sends the Auth Res including a code “STATUS_NOT_COMPATIBLE” to the terminal **100** in T422. This code is a code indicating that the roles are in conflict. In other words, in the present case, the DPP process between the terminal **100** and the printer **10** fails due to failure in role assignment between the terminal **100** and the printer **10**.

(87) When the printer **10** sends the Auth Res to the terminal **100** in T422 (i.e., when the DPP process fails), the printer **10** shifts to the G/O state in T430. In the Wi-Fi network formed by the printer **10** which operates in the G/O state, a character string matching the above code “STATUS_NOT_COMPATIBLE” is used as an SSID.

(88) When the terminal **100** receives the Auth Res from the printer **10** in T422, (i.e., when the DPP process fails), the terminal **100** broadcasts the Probe Req in T432.

(89) When the printer **10** receives the Probe Req from the terminal **100** in T432, the printer **10** sends the Probe Res to the terminal **100** in T434. The Probe Res includes the SSID “STATUS_NOT_COMPATIBLE”. As described above, since the printer **10** sends the Probe Res including the code “STATUS_NOT_COMPATIBLE” to the terminal **100**, the printer **10** can send, to the terminal **100**, information indicating that the DPP process has failed.

(90) When the terminal **100** receives the Probe Res from the printer **10** in T434, the terminal **100** sends the G/O stop instruction to the printer **10** in T436. Consequently, the printer **10** stops the G/O state in T438. Consequently, the Wi-Fi network formed by the printer **10** disappears. Thus, process load on the printer **10** can be reduced.

(91) When the terminal **100** receives the Probe Res from the printer **10** in T434, the terminal **100** causes the display unit **114** to display a notification screen SC4 in T440. The notification screen SC4 includes a message indicating that the DPP process with the printer **10** has failed and the code “STATUS_NOT_COMPATIBLE” included in the Probe Res. By viewing the notification screen SC4, the user can acknowledge that the DPP process with the printer **10** has failed, and the cause of the failure (i.e., conflict in roles).

Effects of First Embodiment

(92) According to the above configuration, when the Wi-Fi connection between the printer **10** and the AP **6a** is established (see T150 of FIG. 3), the printer **10** sends, to the terminal **100**, the Probe Res including the IP address IPpr of the printer **10** as the SSID (T164 of FIG. 4). Consequently, the terminal **100** can execute the connectivity confirmation by using the IP address IPpr. Thus, user convenience is enhanced.

(93) (Corresponding Relationships)

(94) The printer **10**, the terminal **100**, the AP **6a** and the AP **6b** are an example of “communication device”, “terminal device”, “first access point (and access point)” and “second access point”, respectively. The display unit **14** of the printer **10** is an example of “output unit”. The Wi-Fi I/F **16** and the Wi-Fi I/F **116** are an example of “Wi-Fi interface” of “communication device” and “Wi-Fi interface” of “terminal device”, respectively. The public key PKpr of the printer **10** and the PIN code “1234” are an example of “bootstrapping key” and “authentication information”, respectively. The QR Codes QR1 is an example of “outputted information (and code image)”.

(95) The Auth Req and the Auth Res are an example of “authentication request” and “authentication response”, respectively. The Probe Req and the Probe Res are an example of “Probe request” and “Probe response”, respectively. The CO for printer is an example of “first connection information”. The SSID “S2” and the PW “P2” of the AP 6b are an example of “second connection information”. The IP address IPpr is an example of “confirmation information”. The code “STATUS_NOT_COMPATIBLE” is an example of “error information”. The G/O state and the state of not operating as the parent station of the Wi-Fi network are an example of “parent state” and “non-parent state”, respectively. The SSID used when the printer 10 operates in the G/O state is an example of “network identification information”. The Wi-Fi connection established in T150 of FIG. 3 and the WFD connection established in T272 of FIG. 5 are an example of “first Wi-Fi connection” and “second Wi-Fi connection”, respectively. The WFD connection establishment request of T270 of FIG. 5 is an example of “establishment request”. The G/O stop instruction sent in T298 of FIG. 6 is an example of “parent state stop instruction”. The message indicating an approach to connect to another AP (e.g., the AP 6b) included in the notification screen SC2 is an example of “approach information”.

(96) The process of T104 of FIG. 3 is an example of “cause the output unit to output information” executed by “communication device”. The processes of T120 to T132 of FIG. 3 are examples of “execute a Device Provisioning Protocol (DPP) process” executed by “communication device”. In particular, the process of T120, the process of T122 and the process of T132 are an example of “receiving an authentication request”, “sending an authentication response to the terminal device” and “receiving first connection information” executed by “communication device”, respectively. The process of T160 of FIG. 4 (and the process of T430 of FIG. 8), the process of T164 and the process of T172 are an example of “in the case where the first Wi-Fi connection between the communication device and the first access point is established, shift a state of the communication device from a non-parent state to a parent state”, “send confirmation information” and “in a case where a predetermined time elapses without the establishment request being received from the terminal device after the connectivity confirmation signal has been received from the terminal device via the Wi-Fi interface, shift the state of the communication device from the parent state to the non-parent state” executed by “communication device”, respectively. The process of T270, the process of T272, the process of T274 and the process of T276 of FIG. 5 are an example of “in a case where the connectivity confirmation fails at the terminal device, receive an establishment request”, “establish a second Wi-Fi connection”, “receive an inquiry” and “send a response” executed by “communication device”, respectively. The process of T290 and the process of T299 of FIG. 6 are an example of “receive second connection information” and “in a case where a parent state stop instruction is received from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, shift the state of the communication device from the parent state to the non-parent state” executed by “communication device”, respectively. The process of T434 of FIG. 8 is an example of “send error information” executed by “communication device”.

(97) The processes of T120 to T132 of FIG. 3 are examples of “execute a Device Provisioning Protocol (DPP) process” executed by “application program”. In particular, the process of T120, the process of T122 and the process of T132 are examples of “sending an authentication request”, “receiving an authentication response” and “sending connection information” executed by “application program”, respectively. The process of T162 and the process of T164 are an example of “send a Probe request” and “receive confirmation information” executed by “application program”, respectively. The processes of T166 and T168 of FIG. 4 are example of “execute the connectivity confirmation” executed by “application program”. The process of T270, the process of T272, the process of T274, the process of T276 and the process of T278 of FIG. 5 are an example of “in a case where the connectivity confirmation fails, send an establishment request”, “establish a second Wi-Fi connection”, “send an inquiry”, “receive a response” and “cause a display unit of the terminal device to display approach information” executed by “application program”, respectively.

Second Embodiment

(98) Next, Second Embodiment will be described. The second embodiment is different from the first embodiment in its method to send the IP address IPpr of the printer **10** (or the code “STATUS_NOT_COMPATIBLE”) to the terminal **100**. In the second embodiment, the QR Code adhered to the housing **50** of the printer **10** is a QR Code QR2. The QR Code QR2 is a code image obtained by coding respective pieces of information included in the QR Code QR1 (i.e., the public key PKpr, the MAC address MAC pr2 and the PIN code “1234”) and a predetermined SSID “S3”. The SSID “S3” is an SSID used in the Wi-Fi network formed by the printer **10** which operates in the G/O state.

(99) (Case E; FIG. 9)

(100) Case E will firstly be described with reference to FIG. 9. Case E is a case in which the connectivity confirmation succeeds. FIG. 9 is continuation of FIG. 3. In other words, in the initial state of FIG. 9, the Wi-Fi connection is established between the printer **10** and the AP **6a**. In T104 of FIG. 3 before FIG. 9, the QR Code QR2 is displayed instead of the QR Code QR1.

(101) The processes of T560 to T564 are the same as the processes of T160 to T164 of FIG. 4 except that the SSID to be used is “S3”. The processes of T570 and T572 are the same as the processes of T270 and T272 of FIG. 5.

(102) In T574, the terminal **100** sends a DPP success confirmation to the printer **10** by using the WFD connection.

(103) Since the DPP process between the terminal **100** and the printer **10** has already succeeded, when the printer **10** receives the DPP success confirmation from the terminal **100** in T574, the printer **10** sends a signal indicating that the DPP process has succeeded to the terminal **100** by using the WFD connection in T576. This signal includes the IP address IPpr of the printer **10**. As described above, the printer **10** can send, to the terminal **100**, the IP address IPpr assigned by the AP **6a**.

(104) When the terminal **100** receives the signal including the IP address IPpr from the printer **10** in T576, the terminal **100** sends a connectivity confirmation signal including the received IP address IPpr as a recipient address in T580. The processes of T580 and T582 are the same as the processes of T166 and T168 of FIG. 4. The processes of T584 and T586 are the same as the processes of T298 and T299 of FIG. 6. The process of T588 is the same as the process of T170 of FIG. 4. As with Case A of the first embodiment, the user can acknowledge that the terminal **100** and the printer **10** can communicate with each other via the AP **6a** by viewing the notification screen SC1.

(105) (Case F; FIG. 10)

(106) Next, Case F will be described with reference to FIG. 10. Case F is a case in which the Wi-Fi connection between the printer **10** and the AP **6a** is established but the connectivity confirmation fails. The initial state of FIG. 10 is the same as the initial state of FIG. 9. In Case F, the same processes as the processes of T560 to T576 of FIG. 9 are executed.

(107) In T678, the terminal **100** sends, to the printer **10**, the connectivity confirmation signal including the IP address IPpr of the printer **10** as a recipient address. In the present case, however, the AP **6a** does not receive the connectivity confirmation signal because of the weak radio environment around the AP **6a**. Consequently, the printer **10** does not receive the connectivity confirmation signal from the terminal **100**. In other words, in the present case, the terminal **100** does not receive the connectivity OK signal even when a predetermined time elapses from when the connectivity confirmation signal was sent, thus the terminal **100** determines that the connectivity confirmation has failed.

(108) The processes of T684 to T688 are the same as the processes of T274 to T278 of FIG. 5. Consequently, the user can acknowledge what sort of approach should be taken to execute the Wi-Fi communication with the printer **10** (i.e., connect with another AP) by viewing the notification screen SC2. Thereafter, the processes as in FIG. 6 are executed, by which the terminal **100** and the printer **10** can communicate with each other via the AP **6b**.

(109) (Case G; FIG. 11)

(110) Next, Case G will be described with reference to FIG. 11. Case G is a case in which the DPP process between the terminal 100 and the printer 10 fails. The initial state of FIG. 11 is the same as the initial state of FIG. 8 except that the QR Code adhered to the housing of the printer 10 is QR2.

(111) In Case G, the same processes as the processes of T400 to T422 of FIG. 8 are firstly executed. In other words, in Case G as well, the DPP process between the terminal 100 and the printer 10 fails. The processes of T760 to T764 are the same as the processes of T430 to T434 of FIG. 8 except that the SSID to be used is “S3”. The processes of T770 to T774 are the same as the processes of T570 to T574 of FIG. 9.

(112) Since the DPP process between the terminal 100 and the printer 10 has failed (see FIG. 8), when the printer 10 receives the DPP success confirmation from the terminal 100 in T774, the printer 10 sends, to the terminal 100, a signal indicating that the DPP process has failed in T776 by using the established WFD connection. This signal includes the code

“STATUS_NOT_COMPATIBLE” (see T422 of FIG. 8). As described above, the printer 10 can send the information indicating that the DPP process has failed to the terminal 100. The processes of T778 to T780 are the same as the processes of T436 to T440 of FIG. 8. In the present case as well, the user can acknowledge that the DPP process with the printer 10 has failed and the cause of the failure (i.e., conflict in roles) by viewing the notification screen SC4.

Effects of Second Embodiment

(113) According to the above configuration, when the Wi-Fi connection between the printer 10 and the AP 6a is established (see T150 of FIG. 3), the printer 10 sends the signal including the IP address IPpr of the printer 10 to the terminal 100 (T576 of FIG. 9). Consequently, the terminal 100 can execute the connectivity confirmation by using the IP address IPpr. Thus, user convenience is enhanced.

(114) (Corresponding Relationships)

(115) The process of T560, the process of T570, the process of T572 (and the process of T772 of FIG. 11), the process of T576 and the process of T586 of FIG. 9 are an example of “in the case where the first Wi-Fi connection between the communication device and the first access point is established, shift a state of the communication device from a non-parent state to a parent state”, “after the state of the communication device has been shifted to the parent state, receive an establishment request”, “establish a second Wi-Fi connection”, “send confirmation information”, and “in a case where a parent state stop instruction is received from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, shift the state of the communication device from the parent state to the non-parent state” executed by “communication device”, respectively. The process of T684 of FIG. 10 and the process of T686 of FIG. 10 are an example of “receive an inquiry” and “send a response” executed by “communication device”, respectively. The process of T760, the process of T770 and the process of T776 of FIG. 11 are an example of “in the case where the DPP process fails, shift a state of the communication device from a non-parent state to a parent state”, “in a case where the state of the communication device is shifted to the parent state, receive an establishment request” and “send error information”, respectively.

(116) The process of T570, the process of T572, the process of T576 and the processes of T580 and T582 of FIG. 9 are an example of “in a case where the first Wi-Fi connection between the communication device and the access point is established, send an establishment request”, “establish a second Wi-Fi connection”, “receive confirmation information” and “execute the connectivity confirmation” executed by “application program”, respectively. The process of T684, the process of T686 and the process of T688 of FIG. 10 are an example of “send an inquiry”, “receive a response”, “cause a display unit of the terminal device to display approach information” executed by “application program”, respectively.

(117) While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations,

improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

(118) (Modification 1) In T160 of FIG. 4, the printer **10** may shift to the G/O state, and in the Wi-Fi network formed by the printer **10** which operates in the G/O state, a Universally Unique Identifier (UUID) of the printer **10** may be used as an SSID. In this case, in T164, the printer **10** may send the Probe Res including the UUID to the terminal **100**. Then, the terminal **100** may execute the connectivity confirmation by using this UUID. In the present modification, the UUID of the printer **10** is an example of “confirmation information”.

(119) (Modification 2) A timing at which the process of T160 of FIG. 4 is executed is not limited to the aspect of the embodiment. For example, the printer **10** may shift to the G/O state before sending the Status Query Result to the terminal **100** (see T156 of FIG. 3). In other words, the timing at which the process of T160 is executed only needs to be after the IP address IPpr is stored in the printer **10** (i.e., after T154 of FIG. 3). Moreover, the timing at which the process of T560 of FIG. 9 is executed only needs to be after the Wi-Fi connection between the printer **10** and the AP **6a** is established (i.e., after T150 of FIG. 3).

(120) (Modification 3) In T160, the printer **10** may activate the SoftAP instead of shifting to the G/O state. In this case, the IP address IPpr assigned to the printer **10** may be used as an SSID identifying a wireless network formed by the SoftAP.

(121) (Modification 4) The terminal **100** can omit the processes from T270 of FIG. 5. In other words, the terminal **100** may not send the inquiry to the printer **10** when the connectivity confirmation fails. In this case, the terminal **100** may cause the display unit **14** to display a screen including only a message indicating that the communication with the printer **10** via the AP **6a** has failed. In the present modification, “receive an inquiry”, “send a response” and “receive second connection information” executed by “communication device” can be omitted. Further, “send an inquiry”, “receive a response” and “cause a display unit of the terminal device to display approach information” executed by “application program” can be omitted.

(122) (Modification 5) The terminal **100** can omit the processes of T298 and T299 of FIG. 6. In the present modification, “in a case where a parent state stop instruction is received from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, shift the state of the communication device from the parent state to the non-parent state” executed by “communication device” can be omitted. In another modification, instead of the process of T298, the printer **10** may execute the process of T299 when the printer **10** directly obtains the G/O stop instruction from the user via the operation unit **12** of the printer **10**.

(123) (Modification 6) In T104 of FIG. 3, the printer **10** may cause the print executing unit **18** to print the QR Code QR1. In the present modification, the print executing unit **18** is an example of “output unit” and printing of the QR Code QR1 is an example of “cause the output unit to output information” executed by “communication device”. In another modification, in T104, the printer **10** may supply the public key PKpr, the MAC address MACpr2 and the PIN Code to a Near Field Communication (NFC) I/F (illustration omitted). In this case, the terminal **100** may obtain information such as the public key PKpr via NFC communication in response to the terminal **100** being brought closer to the printer **10**. In the present modification, the information such as the public key PKpr is an example of “outputted information”, the NFC I/F is an example of “output unit”, and supply of the public key PKpr and the like to the NFC I/F is an example of “cause the output unit to output information” executed by “communication device”. In another modification,

in T104, the printer **10** may supply the public key PKpr, the MAC address MACpr2 and the PIN Code to a Bluetooth (BT) I/F (illustration omitted). Bluetooth is a registered trademark of Bluetooth SIG. In this case, the terminal **100** may obtain the information such as the public key PKpr via BT communication. In the present modification, the BT I/F is an example of “output unit”, supply of the public key PKpr and the like to the BT I/F is an example of “cause the output unit to output information” executed by “communication device”.

(124) (Modification 7) The QR Code QR1 may be a code image obtained by coding information not including the PIN Code “1234”. In this case, the user may obtain the PIN code “1234” from the printer **10** by a method other than capturing of the QR Code QR1. In another modification, the PIN code “1234” may not be used to establish the WFD connection between the terminal **100** and the printer **10**. In this case, the printer **10** may not store the PIN code “1234”.

(125) (Modification 8) The terminal **100** and the printer **10** may omit the processes from T430 of FIG. **8**. In the present modification, “send error information” executed by “communication device” can be omitted. In the above embodiment, the code “STATUS_NOT_COMPATIBLE” was sent in response to the failure of the role assignment in the DPP process, however, when the DPP process fails due to a different cause in the DPP process, the printer **10** may send a code indicating the cause to the terminal **100**. In the present modification, the code indicating the cause is an example of “error information”.

(126) (Modification 9) In the above embodiments, the processes of FIGS. **2** to **11** are implemented by a software such as the program **36**, however, at least one of these processes may be implemented by hardware such as a logic circuit.

Claims

1. A communication device, comprising: a Wi-Fi interface configured to execute Wi-Fi communication according to a Wi-Fi standard; and a controller, wherein the controller is configured to: execute a Device Provisioning Protocol (DPP) process according to the DPP of the Wi-Fi standard, wherein the DPP process comprises: receiving an authentication request in which a bootstrapping key of the communication device is used from a terminal device via the Wi-Fi interface; in a case where the authentication request is received from the terminal device, sending an authentication response to the terminal device via the Wi-Fi interface; and after the authentication response has been sent to the terminal device, receiving first connection information from the terminal device via the Wi-Fi interface, wherein the first connection information is used by the communication device for establishing a first Wi-Fi connection between the communication device and a first access point via the Wi-Fi interface, and in a case where the first Wi-Fi connection between the communication device and the first access point is established, send confirmation information to the terminal device via the Wi-Fi interface, wherein the confirmation information is used by the terminal device for executing connectivity confirmation to confirm whether the Wi-Fi communication between the communication device and the terminal device via the first access point is executable.

2. The communication device as in claim 1, wherein the controller is further configured to: in the case where the first Wi-Fi connection between the communication device and the first access point is established, shift a state of the communication device from a non-parent state to a parent state, wherein the non-parent state is a state in which the communication device does not operate as a parent station of a wireless network, the parent state is a state in which the communication device operates as the parent station of the wireless network, and network identification information for identifying the wireless network includes the confirmation information, wherein in a case where a Probe request is received from the terminal device via the Wi-Fi interface, the controller is configured to send the confirmation information to the terminal device by sending, to the terminal device, a Probe response including the network identification information via the Wi-Fi interface.

3. The communication device as in claim 2, wherein the controller is further configured to: in a case where the connectivity confirmation fails at the terminal device, receive an establishment request from the terminal device via the Wi-Fi interface; and in a case where the establishment request is received from the terminal device, establish a second Wi-Fi connection between the communication device and the terminal device via the Wi-Fi interface so as to join the terminal device to the wireless network as a child station.
4. The communication device as in claim 3, wherein the connectivity confirmation includes a connectivity confirmation signal being sent from the terminal device to the communication device via the first access point, wherein the controller is further configured to: in a case where a predetermined time elapses without the establishment request being received from the terminal device after the connectivity confirmation signal has been received from the terminal device via the Wi-Fi interface, shift the state of the communication device from the parent state to the non-parent state.
5. The communication device as in claim 3, wherein the controller is further configured to: in the case where the connectivity confirmation fails at the terminal device, receive an inquiry related to the connectivity confirmation from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, and in a case where the inquiry is received from the terminal device, send a response to the inquiry to the terminal device via the Wi-Fi interface by using the second Wi-Fi connection.
6. The communication device as in claim 3, wherein the controller is further configured to: in the case where the connectivity confirmation fails at the terminal device, receive second connection information according to a scheme different from the DPP from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, wherein the second connection information is used by the communication device for establishing a third Wi-Fi connection between the communication device and a second access point different from the first access point via the Wi-Fi interface.
7. The communication device as in claim 3, wherein the controller is further configured to: in a case where a parent state stop instruction is received from the terminal device via the Wi-Fi interface by using the second Wi-Fi connection, shift the state of the communication device from the parent state to the non-parent state.
8. The communication device as in claim 3, further comprising an output unit, wherein the controller is further configured to: cause the output unit to output information obtained by using the bootstrapping key and authentication information, wherein in response to the outputted information being obtained by the terminal device, the authentication request is received from the terminal device, and the authentication information is information for the terminal device to establish the second Wi-Fi connection with the communication device.
9. The communication device as in claim 3, further comprising: a housing to which a code image obtained by coding the bootstrapping key and authentication information is adhered, wherein in response to the code image being captured by the terminal device, the authentication request is received from the terminal device, wherein the authentication information is information for the terminal device to establish the second Wi-Fi connection with the communication device.
10. The communication device as in claim 1, wherein the controller is further configured to: in the case where the first Wi-Fi connection between the communication device and the first access point is established, shift a state of the communication device from a non-parent state to a parent state, wherein the non-parent state is a state in which the communication device does not operate as a parent station of a wireless network, and the parent state is a state in which the communication device operates as the parent station of the wireless network; after the state of the communication device has been shifted to the parent state, receive an establishment request from the terminal device via the Wi-Fi interface; and in a case where the establishment request is received from the terminal device, establish a second Wi-Fi connection between the communication device and the terminal device via the Wi-Fi interface so as to join the terminal device to the wireless network as a

child station, wherein the controller is configured to send the confirmation information to the terminal device via the Wi-Fi interface by using the second Wi-Fi connection.

11. The communication device as in claim 1, wherein the confirmation information is an IP address of the communication device assigned by the first access point.

12. The communication device as in claim 1, wherein the controller is further configured to: in a case where the DPP process fails, send error information indicating the DPP process has failed to the terminal device via the Wi-Fi interface.

13. The communication device as in claim 12, wherein the controller is further configured to: in the case where the DPP process fails, shift a state of the communication device from a non-parent state to a parent state, wherein the non-parent state is a state in which the communication device does not operate as a parent station of a wireless network and the parent state is a state in which the communication device operates as the parent station of the wireless network, and network identification information identifying the wireless network includes the error information, wherein in a case where a Probe request is received from the terminal device via the Wi-Fi interface, the controller is configured to send the error information to the terminal device by sending, to the terminal device, a Probe response including the network identification information via the Wi-Fi interface.

14. The communication device as in claim 12, wherein the controller is further configured to: in the case where the DPP process fails, shift a state of the communication device from a non-parent state to a parent state, wherein the non-parent state is a state in which the communication device does not operate as a parent station of a wireless network and the parent state is a state in which the communication device operates as the parent station of the wireless network; in a case where the state of the communication device is shifted to the parent state, receive an establishment request from the terminal device via the Wi-Fi interface; and in a case where the establishment request is received from the terminal device, establish a second Wi-Fi connection between the communication device and the terminal device via the Wi-Fi interface so as to join the terminal device to the wireless network as a child station, wherein the controller is configured to send the error information to the terminal device via the Wi-Fi interface by using the second Wi-Fi connection.

15. A non-transitory computer-readable recording medium storing computer-readable instructions for a communication device, the communication device comprises: a Wi-Fi interface configured to execute Wi-Fi communication according to a Wi-Fi standard; and a processor, wherein the computer-readable instructions, when executed by the processor, causing the communication device to: execute a Device Provisioning Protocol (DPP) process according to the DPP of the Wi-Fi standard, wherein the DPP process comprises: receiving an authentication request in which a bootstrapping key of the communication device is used from a terminal device via the Wi-Fi interface; in a case where the authentication request is received from the terminal device, sending an authentication response to the terminal device via the Wi-Fi interface; and after the authentication response has been sent to the terminal device, receiving first connection information from the terminal device via the Wi-Fi interface, wherein the first connection information is used by the communication device for establishing a first Wi-Fi connection between the communication device and a first access point via the Wi-Fi interface, and in a case where the first Wi-Fi connection between the communication device and the first access point is established, send confirmation information to the terminal device via the Wi-Fi interface, wherein the confirmation information is used by the terminal device for executing connectivity confirmation to confirm whether the Wi-Fi communication between the communication device and the terminal device via the first access point is executable.

16. A non-transitory computer-readable recording medium storing an application program including computer-readable instructions for a terminal device, the terminal device comprises: a Wi-Fi interface configured to execute Wi-Fi communication according to a Wi-Fi standard; and a

processor, wherein the computer-readable instructions, when executed by the processor, causing the terminal device to: execute a Device Provisioning Protocol (DPP) process according to the DPP of the Wi-Fi standard, wherein the DPP process comprises: sending an authentication request in which a bootstrapping key of a communication device is used to the communication device via the Wi-Fi interface; in a case where the authentication request is sent to the communication device, receiving an authentication response from the communication device via the Wi-Fi interface; and after the authentication response has been received from the communication device, sending connection information to the communication device via the Wi-Fi interface, wherein the connection information is used by the communication device for establishing a first Wi-Fi connection between the communication device and an access point via the Wi-Fi interface, in a case where the first Wi-Fi connection between the communication device and the access point is established, receive confirmation information from the communication device via the Wi-Fi interface, wherein the confirmation information is used by the terminal device for executing connectivity confirmation to confirm whether the Wi-Fi communication between the communication device and the terminal device via the access point is executable; and in a case where the confirmation information is received from the communication device, execute the connectivity confirmation by using the confirmation information.

17. The non-transitory computer-readable recording medium as in claim 16, wherein the computer-readable instructions, when executed by the processor, further cause the terminal device to: in the case where the first Wi-Fi connection between the communication device and the access point is established, send a Probe request to the communication device via the Wi-Fi interface, wherein the confirmation information is received from the communication device by receiving, from the communication device operating as a parent station of a wireless network, a Probe response including network identification information including the confirmation information via the Wi-Fi interface, the network identification information being for identifying the wireless network.

18. The non-transitory computer-readable recording medium as in claim 17, wherein the computer-readable instructions, when executed by the processor, further cause the terminal device to: in a case where the connectivity confirmation fails, send an establishment request to the communication device via the Wi-Fi interface; and in a case where the establishment request is sent to the communication device, establish a second Wi-Fi connection between the communication device operating as the parent station of the wireless network and the terminal device via the Wi-Fi interface so as to join the wireless network as a child station.

19. The non-transitory computer-readable recording medium as in claim 16, wherein the computer-readable instructions, when executed by the processor, further cause the terminal device to: in a case where the first Wi-Fi connection between the communication device and the access point is established, send an establishment request to the communication device via the Wi-Fi interface; and in a case where the establishment request is sent to the communication device, establish a second Wi-Fi connection between the communication device operating as a parent station of a wireless network and the terminal device via the Wi-Fi interface so as to join the wireless network as a child station, wherein the confirmation information is received from the communication device via the Wi-Fi interface by using the second Wi-Fi connection.

20. The non-transitory computer-readable recording medium as in claim 18, wherein the computer-readable instructions, when executed by the processor, further cause the terminal device to: in a case where the connectivity confirmation fails, send an inquiry related to the connectivity confirmation to the communication device via the Wi-Fi interface by using the second Wi-Fi connection; in a case where the inquiry is sent to the communication device, receive a response to the inquiry from the communication device via the Wi-Fi interface by using the second Wi-Fi connection; and in a case where the response is received from the communication device, cause a display unit of the terminal device to display approach information according to the response.
